Contents

Supplementary Appendix

Stimulant-Opioid Comorbidity Study, 2015-2024

This supplement reports four post hoc sensitivity analyses and the data dictionary for the derived analysis variables.

Sensitivity Analysis 1: Comparison of 2015-2019 and 2020-2024

Methods

We conducted a post hoc analysis to determine whether the adjusted drug-group coefficients differed between 2015-2019 and 2020-2024. For each of the four binary outcomes, we fit logistic regression models with stimulant-only deaths as the reference group and adjusted for age in 10-year units, sex, race, and Hispanic origin. We estimated period-specific models and tested drug-group-by-period interactions in pooled models. Period interactions were also included for each adjustment covariate, allowing their associations to differ between periods. The exponentiated drug-group interaction is the ratio of adjusted odds ratios (2020-2024 divided by 2015-2019).

Results

The complete-case sample included 309,059 deaths in 2015-2019 and 522,580 deaths in 2020-2024. For all these comparisons the direction and interpretation of the coefficents are consistent. There were some stastical differences between coefficents between the two time peroids. The opioid-only coefficient changed for cardiovascular disease and no additional condition. The opioid-stimulant coefficient changed for other medical conditions. Neither drug-group coefficient changed for cerebrovascular disease. Because the periods were defined by calendar year, these estimates describe temporal differences and do not establish that COVID-19 caused the changes.

eTable S1. Adjusted coefficients by calendar period

Outcome	Group	2015-2019 β (SE)	2015-2019 aOR (95% CI)	2020-2024 β (SE)	2020-2024 aOR (95% CI)	Ratio of aORs (95% CI)	P for change
CV disease	Opioid + stim.	-1.487 (0.014)	0.23 (0.22-0.23)	-1.466 (0.009)	0.23 (0.23-0.23)	1.02 (0.99-1.05)	.181
CV disease	Opioid only	-1.386 (0.011)	0.25 (0.24-0.26)	-1.334 (0.009)	0.26 (0.26-0.27)	1.05 (1.03-1.08)	<.001
CBV disease	Opioid + stim.	-2.321 (0.052)	0.10 (0.09-0.11)	-2.363 (0.033)	0.09 (0.09-0.10)	0.96 (0.85-1.08)	.492
CBV disease	Opioid only	-2.129 (0.034)	0.12 (0.11-0.13)	-2.107 (0.029)	0.12 (0.11-0.13)	1.02 (0.94-1.12)	.625

Outcome	Group	2015-2019 β (SE)	2015-2019 aOR (95% CI)	2020-2024 β (SE)	2020-2024 aOR (95% CI)	Ratio of aORs (95% CI)	P for change
Other medical	Opioid + stim.	-1.207 (0.024)	0.30 (0.29-0.31)	-1.390 (0.015)	0.25 (0.24-0.26)	0.83 (0.79-0.88)	<.001
Other medical	Opioid only	-0.857 (0.016)	0.42 (0.41-0.44)	-0.839 (0.013)	0.43 (0.42-0.44)	1.02 (0.98-1.06)	.368
No additional	Opioid + stim.	1.633 (0.013)	5.12 (4.99-5.25)	1.626 (0.009)	5.08 (5.00-5.17)	0.99 (0.96-1.02)	.674
No additional	Opioid only	1.489 (0.010)	4.43 (4.34-4.53)	1.431 (0.008)	4.18 (4.11-4.25)	0.94 (0.92-0.97)	<.001

Note. aOR indicates adjusted odds ratio; CI, confidence interval; CV, cardiovascular; CBV, cerebrovascular; and stim., stimulant. "Opioid + stim." denotes the opioid-stimulant group. The stimulant-only group was the reference. All models adjusted for age, sex, race, and Hispanic origin. The P value tests the drug-group-by-period interaction. This was a post hoc analysis.

Sensitivity Analysis 2: Cocaine and Other Psychostimulants Analyzed Separately

Methods

We repeated the adjusted logistic regression models separately for cocaine and other psychostimulant involvement. Cocaine involvement was defined by F14, T40.5, or R78.2; other psychostimulant involvement was defined by F15 or T43.6; and opioid involvement was defined by F11, T40.0-T40.4, T40.6, or R78.1. The cocaine analysis excluded deaths involving other psychostimulants, and the other-psychostimulant analysis excluded deaths involving cocaine. Each model compared the opioid-stimulant and opioid-only groups with the corresponding stimulant-only group and adjusted for age in 10-year units, sex, race, and Hispanic origin.

Results

The cocaine analysis included 559,791 deaths: 68,637 cocaine-only, 133,807 opioid-cocaine, and 357,347 opioid-only deaths. The other-psychostimulant analysis included 600,001 deaths: 125,951 other-psychostimulant-only, 116,703 opioid-other-psychostimulant, and 357,347 opioid-only deaths. The direction and interpretaiton of the coefficients is the same between the two drug categories. In both analyses, opioid-involved groups had lower adjusted odds of the medical-condition outcomes and higher adjusted odds of having no additional condition.

eTable S2a. Cocaine

Outcome	Comparison	β (SE)	aOR (95% CI)	P value
Cardiovascular disease	Opioid-cocaine vs Cocaine-only	-1.676 (0.012)	0.19 (0.18-0.19)	<.001
Cardiovascular disease	Opioid-only vs Cocaine-only	-1.397 (0.010)	0.25 (0.24-0.25)	<.001
Cerebrovascular disease	Opioid-cocaine vs Cocaine-only	-2.575 (0.044)	0.08 (0.07-0.08)	<.001
Cerebrovascular disease	Opioid-only vs Cocaine-only	-2.094 (0.027)	0.12 (0.12-0.13)	<.001
Other medical condition	Opioid-cocaine vs Cocaine-only	-1.483 (0.021)	0.23 (0.22-0.24)	<.001
Other medical condition	Opioid-only vs Cocaine-only	-0.701 (0.015)	0.50 (0.48-0.51)	<.001
No additional condition	Opioid-cocaine vs Cocaine-only	1.832 (0.011)	6.25 (6.11-6.39)	<.001
No additional condition	Opioid-only vs Cocaine-only	1.473 (0.010)	4.36 (4.28-4.45)	<.001

Note. aOR indicates adjusted odds ratio; CI, confidence interval. This analysis excluded deaths involving the other stimulant class. All models adjusted for age, sex, race, and Hispanic origin. This was a post hoc analysis.

eTable S2b. Other psychostimulants

Outcome	Comparison	β (SE)	aOR (95% CI)	P value
Cardiovascular disease	Opioid-other psychostimulant vs Other psychostimulant-only	-1.301 (0.010)	0.27 (0.27-0.28)	<.001
Cardiovascular disease	Opioid-only vs Other psychostimulant-only	-1.351 (0.008)	0.26 (0.26-0.26)	<.001
Cerebrovascular disease	Opioid-other psychostimulant vs Other psychostimulant-only	-2.151 (0.039)	0.12 (0.11-0.13)	<.001
Cerebrovascular disease	Opioid-only vs Other psychostimulant-only	-2.113 (0.023)	0.12 (0.12-0.13)	<.001
Other medical condition	Opioid-other psychostimulant vs Other psychostimulant-only	-1.113 (0.017)	0.33 (0.32-0.34)	<.001
Other medical condition	Opioid-only vs Other psychostimulant-only	-0.918 (0.011)	0.40 (0.39-0.41)	<.001

Outcome	Comparison	β (SE)	aOR (95% CI)	P value
No additional condition	Opioid-other psychostimulant vs Other psychostimulant-only	1.453 (0.010)	4.27 (4.19-4.36)	<.001
No additional condition	Opioid-only vs Other psychostimulant-only	1.463 (0.007)	4.32 (4.26-4.38)	<.001

Note. aOR indicates adjusted odds ratio; CI, confidence interval. This analysis excluded deaths involving the other stimulant class. All models adjusted for age, sex, race, and Hispanic origin. This was a post hoc analysis.

Sensitivity Analysis 3: Inclusive and Restricted Drug-Involvement Definitions

Methods

We repeated the adjusted logistic regression models using a restricted drug-involvement definition based only on poisoning codes (T40.0-T40.6, with T40.5 classified as stimulant rather than opioid involvement; and T43.6) and toxicologic blood-finding codes (R78.1 and R78.2). The restricted definition did not include opioid (F11), cocaine (F14), or other-stimulant (F15) mental and behavioral disorder codes. Deaths were reassigned to the three mutually exclusive drug groups; deaths without a qualifying poisoning or blood-finding code were excluded. Maternal-exposure codes were not included in either definition. Outcomes, covariates, study years, and the stimulant-only reference group were unchanged.

Results

The restricted definition included 756,580 deaths, compared with 831,852 under the inclusive definition. It retained the direction and statistical significance of every primary drug-group association. Compared with the inclusive estimates, the restricted estimates were generally closer to the null for cardiovascular, cerebrovascular, and other medical conditions and remained strongly positive for no additional condition.

eTable S3a. ICD-10 drug-involvement codes by analytic definition

Drug class	ICD-10 code	Description	Definition status
Opioid	F11 (all subcodes)	Opioid-related mental and behavioral disorders	Inclusive only
Opioid	T40.0	Poisoning by opium	Inclusive and restricted
Opioid	T40.1	Poisoning by heroin	Inclusive and restricted
Opioid	T40.2	Poisoning by other opioids	Inclusive and restricted
Opioid	T40.3	Poisoning by methadone	Inclusive and restricted
Opioid	T40.4	Poisoning by other synthetic narcotics	Inclusive and restricted
Opioid	T40.6	Poisoning by other and unspecified narcotics	Inclusive and restricted
Opioid	R78.1	Finding of opiate drug in blood	Inclusive and restricted
Stimulant-cocaine	F14 (all subcodes)	Cocaine-related mental and behavioral disorders	Inclusive only
Stimulant-other	F15 (all subcodes)	Other-stimulant-related mental and behavioral disorders	Inclusive only
Stimulant-cocaine	T40.5	Poisoning by cocaine	Inclusive and restricted

Drug class	ICD-10 code	Description	Definition status
Stimulant-other	T43.6	Poisoning by psychostimulants with abuse potential	Inclusive and restricted
Stimulant-cocaine	R78.2	Finding of cocaine in blood	Inclusive and restricted
Excluded	P04.14	Newborn affected by maternal use of opiates	Excluded from both
Excluded	P04.16	Newborn affected by maternal use of amphetamines	Excluded from both
Excluded	P04.41	Newborn affected by maternal use of cocaine	Excluded from both

Note. T40.5 was classified as stimulant involvement and was excluded from opioid involvement. Codes P04.14, P04.16, and P04.41 are shown to document the team's decision to exclude maternal-exposure codes from both definitions. ICD-10 indicates International Statistical Classification of Diseases and Related Health Problems, Tenth Revision.

eTable S3b. Adjusted estimates under inclusive and restricted definitions

Outcome	Group	Inclusive β (SE)	Inclusive aOR (95% CI)	Restricted β (SE)	Restricted aOR (95% CI)
CV disease	Opioid + stim.	-1.481 (0.007)	0.23 (0.22-0.23)	-1.292 (0.008)	0.27 (0.27-0.28)
CV disease	Opioid only	-1.354 (0.007)	0.26 (0.25-0.26)	-1.200 (0.007)	0.30 (0.30-0.31)
CBV disease	Opioid + stim.	-2.362 (0.028)	0.09 (0.09-0.10)	-2.141 (0.032)	0.12 (0.11-0.13)
CBV disease	Opioid only	-2.120 (0.022)	0.12 (0.11-0.13)	-2.062 (0.027)	0.13 (0.12-0.13)
Other medical	Opioid + stim.	-1.342 (0.013)	0.26 (0.25-0.27)	-1.080 (0.015)	0.34 (0.33-0.35)
Other medical	Opioid only	-0.853 (0.010)	0.43 (0.42-0.43)	-0.783 (0.012)	0.46 (0.45-0.47)
No additional	Opioid + stim.	1.638 (0.007)	5.15 (5.07-5.22)	1.394 (0.008)	4.03 (3.97-4.09)
No additional	Opioid only	1.453 (0.007)	4.28 (4.22-4.33)	1.269 (0.007)	3.56 (3.51-3.61)

Note. aOR indicates adjusted odds ratio; CI, confidence interval; CV, cardiovascular; CBV, cerebrovascular; and stim., stimulant. "Opioid + stim." denotes the opioid-stimulant group. The inclusive sample contained 831,852 deaths (model N = 831,639); the restricted sample contained 756,580 deaths (model N = 756,381). The stimulant-only group was the reference. All P values were <.001. This was a post hoc analysis.

Sensitivity Analysis 4: Adjustment for Alcohol Involvement

Methods

We conducted a post hoc sensitivity analysis to determine whether adjustment for alcohol involvement changed the primary drug-group coefficients. Alcohol involvement was defined as at least one alcohol-related mental and behavioral disorder (`F10`), alcoholic liver disease (`K70`), accidental alcohol poisoning or exposure (`X45`), or toxic effect of alcohol (`T51`) code recorded as the underlying cause or in an available record-axis multiple-cause field. We repeated the primary logistic regression models after adding this binary indicator to the adjustment variables. The models included age in 10-year units, sex, race, Hispanic origin, and alcohol involvement. Stimulant-only deaths remained the reference group. Primary and alcohol-adjusted models were fit to the same complete-case sample for each outcome.

Results

The complete-case sample included 831,639 deaths, of which 15.6% had a qualifying alcohol-involvement code. Adding alcohol involvement changed the drug-group coefficients by 0.002 to 0.019 in absolute value. The largest proportional change in coefficient magnitude was 2.1%. No association changed direction or statistical interpretation.

eTable S4. Primary and alcohol-adjusted drug-group coefficients

Outcome	Group	Primary β (SE)	Primary aOR (95% CI)	Alcohol-adjusted β (SE)	Alcohol-adjusted aOR (95% CI)	Change in β
CV disease	Opioid + stim.	-1.481 (0.007)	0.23 (0.22-0.23)	-1.476 (0.007)	0.23 (0.23-0.23)	+0.005
CV disease	Opioid only	-1.354 (0.007)	0.26 (0.25-0.26)	-1.346 (0.007)	0.26 (0.26-0.26)	+0.008
CBV disease	Opioid + stim.	-2.362 (0.028)	0.09 (0.09-0.10)	-2.348 (0.028)	0.10 (0.09-0.10)	+0.014
CBV disease	Opioid only	-2.120 (0.022)	0.12 (0.11-0.13)	-2.101 (0.022)	0.12 (0.12-0.13)	+0.019
Other medical	Opioid + stim.	-1.342 (0.013)	0.26 (0.25-0.27)	-1.354 (0.013)	0.26 (0.25-0.26)	-0.012
Other medical	Opioid only	-0.853 (0.010)	0.43 (0.42-0.43)	-0.871 (0.010)	0.42 (0.41-0.43)	-0.018

Outcome	Group	Primary β (SE)	Primary aOR (95% CI)	Alcohol-adjusted β (SE)	Alcohol-adjusted aOR (95% CI)	Change in β
No additional	Opioid + stim.	1.638 (0.007)	5.15 (5.07-5.22)	1.636 (0.007)	5.13 (5.06-5.21)	-0.002
No additional	Opioid only	1.453 (0.007)	4.28 (4.22-4.33)	1.449 (0.007)	4.26 (4.21-4.32)	-0.004

Note. aOR indicates adjusted odds ratio; CI, confidence interval; CV, cardiovascular; CBV, cerebrovascular; and stim., stimulant. "Opioid + stim." denotes the opioid-stimulant group. Change in β is the alcohol-adjusted coefficient minus the primary coefficient. All models used the same 831,639 complete cases and converged. The stimulant-only group was the reference. All drug-group coefficients in both specifications had P < .001. This was a post hoc analysis.

Alcohol involvement was defined from conditions recorded on the death certificate and should not be interpreted as a pre-exposure confounder. This analysis tests whether the primary drug-group estimates are sensitive to statistical adjustment for co-recorded alcohol involvement.

Data Dictionary

Script 01 creates a local death-record-level RDS file with one row per death for all cases meeting the inclusive opioid or stimulant definition. The repository does not include mortality microdata. Researchers must obtain the annual National Center for Health Statistics Multiple Cause of Death files from the CDC or another authorized source. The NBER mortality-data documentation describes the annual source files and field coding.

How ICD-10 categories are matched

The analysis removes punctuation and spaces from each ICD-10 value before matching. A category shown below without a decimal point includes every descendant subcode because the program matches the beginning of the normalized code. For example, `C50` includes `C50.0` through `C50.9`; `C00-C97` includes all malignant-neoplasm categories and their descendant subcodes. A single subcode such as `B21.2` includes only that subcode and any values recorded beneath it. The program searches the underlying cause and every available record-axis multiple-cause field (`recaxis1` through `recaxis20`).

The tables below reproduce the rules in `scripts/01_build_analysis_data.R`. They are the auditable specification of the implemented definitions, not a general list of every ICD-10 code that could describe a condition.

Identification and exposure variables

Variable	Meaning
`year`	Calendar year of death, 2015-2024.
`comparison_group`	Three mutually exclusive groups under the inclusive definition: `stimulant_no_opiate`, `opiate_stimulant`, or `opiate_no_stimulant`.
`restricted_group`	The same three groups reconstructed from poisoning and toxicologic blood-finding codes only. Missing means the death had an inclusive F11/F14/F15 code but no qualifying restricted code.
`opioid`	At least one F11, T40.0-T40.4, T40.6, or R78.1 code was recorded.
`cocaine`	At least one F14, T40.5, or R78.2 code was recorded.
`psychostimulant`	At least one F15 or T43.6 code was recorded.

Drug-involvement code hierarchy

Drug class	Inclusive definition	Restricted definition	Notes
Opioid	`F11` (mental and behavioral disorders due to opioid use); `T40.0`–`T40.4` and `T40.6` (poisoning by, adverse effect of, or underdosing of the listed narcotics and related substances); `R78.1` (finding of opiate drug in blood).	`T40.0`–`T40.4`, `T40.6`, and `R78.1`.	`T40.5` is excluded from opioid involvement and assigned to stimulant involvement.
Cocaine	`F14` (mental and behavioral disorders due to cocaine use); `T40.5` (cocaine); `R78.2` (finding of cocaine in blood).	`T40.5` and `R78.2`.	Cocaine is one component of the broad stimulant definition.
Other psychostimulant	`F15` (mental and behavioral disorders due to other stimulant use, including caffeine); `T43.6` (psychostimulants with abuse potential).	`T43.6`.	This component is analyzed separately from cocaine in Sensitivity Analysis 2.

The inclusive definition counts all descendant subcodes under `F11`, `F14`, and `F15`, not only selected fourth-character diagnoses. The restricted definition excludes every `F11`, `F14`, and `F15` code. Maternal-exposure codes `P04.14`, `P04.16`, and

P04.41 are excluded from both definitions.

Demographic variables

Variable	Meaning
age_years	NCHS detail age converted to years; values outside 0-125 are missing.
sex	Male or Female after harmonizing numeric and character annual codes.
race4	White, Black, Asian/Pacific Islander (API), or Other. Other includes AIAN, multiracial, and unknown values.
hisp3	Non-Hispanic, Hispanic, or Unknown. Hispanic origin is separate from race.

Cause-of-death condition indicators

Each variable below equals 1 when at least one qualifying ICD-10 code appears as the underlying cause or in an available record-axis field and 0 otherwise. These indicators describe conditions recorded on the death certificate. They do not represent a complete medical history, establish disease onset, or imply that drug involvement caused the condition.

Condition indicator	ICD-10 categories implemented in the code	Category hierarchy represented
Alcohol involvement	F10; K70; X45; T51	Mental and behavioral disorders due to alcohol use; alcoholic liver disease; accidental poisoning by and exposure to alcohol; toxic effect of alcohol. All descendant subcodes are included.

Condition indicator	ICD-10 categories implemented in the code	Category hierarchy represented
Cardiac disease and hypertension	`I00-I09`; `I10-I13`; `I15`; `I20-I22`; `I24-I28`; `I30-I31`; `I33-I38`; `I40`; `I42-I51`; `I70-I71`	Rheumatic heart disease; selected hypertensive disease; ischemic heart disease; pulmonary heart disease and pulmonary-circulation disorders; selected pericardial, endocardial, valvular, myocardial, conduction, rhythm, heart-failure, and other heart-disease categories; atherosclerosis; aortic aneurysm and dissection. The definition does **not** include every code in Chapter IX. In particular, `I14`, `I16-I19`, `I23`, `I29`, `I32`, `I39`, `I41`, `I52-I59`, and `I72-I99` are not matched by this indicator.

Condition indicator	ICD-10 categories implemented in the code	Category hierarchy represented
Cancer	`B21.2`; `B21.8`; `C00–C97`	HIV disease resulting in other non-Hodgkin lymphoma (`B21.2`) or other malignant neoplasms (`B21.8`), plus the full malignant-neoplasm block: lip/oral cavity/pharynx (`C00–C14`); digestive organs (`C15–C26`); respiratory and intrathoracic organs (`C30–C39`); bone and articular cartilage (`C40–C41`); melanoma and other skin malignancies (`C43–C44`); mesothelial and soft tissue (`C45–C49`); breast (`C50`); female genital organs (`C51–C58`); male genital organs (`C60–C63`); urinary tract (`C64–C68`); eye, brain, and other central nervous system (`C69–C72`); thyroid and other endocrine glands (`C73–C75`); ill-defined, secondary, and unspecified sites (`C76–C80`); lymphoid, hematopoietic, and related tissue (`C81–C96`); and independent multiple-site malignancies (`C97`).

Condition indicator	ICD-10 categories implemented in the code	Category hierarchy represented
Cerebrovascular disease	`I60-I69`	The full cerebrovascular-disease block, including nontraumatic hemorrhage, cerebral infarction, other cerebrovascular diseases, and sequelae.
Diabetes	`E10-E14`	Type 1, type 2, malnutrition-related, other specified, and unspecified diabetes mellitus, including all descendant complication subcodes. No other endocrine, nutritional, or metabolic categories are included.
Hepatobiliary and pancreatic disease	`K70`; `K72-K74`; `K76`; `K80-K82`; `K85-K86`	Alcoholic liver disease; hepatic failure; chronic hepatitis; fibrosis/cirrhosis; other liver disease; cholelithiasis, cholecystitis, and other gallbladder disease; acute pancreatitis and other pancreatic disease. `K71`, `K75`, `K77-K79`, `K83-K84`, and `K87-K93` are not included.

Condition indicator	ICD-10 categories implemented in the code	Category hierarchy represented
Renal disease	`N00-N07`; `N17-N19`; `N25-N28`	Glomerular diseases; acute kidney failure and chronic kidney disease; unspecified kidney failure; disorders resulting from impaired renal tubular function; contracted kidney; small kidney of unknown cause; and other kidney/ureter disorders. `N08-N16`, `N20-N24`, and `N29-N99` are not included.
Viral hepatitis	`B15-B19`	The full viral-hepatitis block: hepatitis A, B, other acute viral hepatitis, chronic viral hepatitis, and unspecified viral hepatitis.
HIV/AIDS	`B20-B24`	The full human immunodeficiency virus disease block.

Condition indicator	ICD-10 categories implemented in the code	Category hierarchy represented
Substance-use or mental-health disorder	`F10–F19`	Mental and behavioral disorders due to psychoactive substance use: alcohol, opioids, cannabinoids, sedatives/hypnotics, cocaine, other stimulants, hallucinogens, tobacco, volatile solvents, and multiple/other substances. This descriptive indicator includes the same `F11`, `F14`, and `F15` families used in the inclusive drug definition.

Condition indicator	ICD-10 categories implemented in the code	Category hierarchy represented
Other accidental injury	`V00-V99`; `W00-W99`; `X00-X39`; `X60-X99`; `Y00-Y09`; `Y15-Y99`	Transport accidents; other external causes of accidental injury; exposure to smoke, fire, flames, heat, hot substances, forces of nature, and other specified accidental threats; intentional self-harm and assault categories; events of undetermined intent outside `Y10-Y14`; legal intervention/war; complications of medical and surgical care; and sequelae/supplementary external-cause factors. After code matching, the indicator is reset to 0 when the NCHS manner-of-death field identifies suicide or homicide. The implemented pattern excludes all `X40-X59` and `Y10-Y14`, not only drug-poisoning categories.

The external-injury indicator is named for its analytic purpose, but the last row states the exact implemented rule. Readers should use those ranges, together with the manner-of-death override, when assessing whether the definition includes the external causes they expect.

R variable names for the condition indicators

- Alcohol involvement: `alcohol_any`
- Cardiac disease and hypertension: `cardiac_hypertension`
- Cancer: `cancer`
- Cerebrovascular disease: `cerebrovascular`
- Diabetes: `diabetes_metabolic`
- Hepatobiliary and pancreatic disease: `hepatobiliary_pancreatic`
- Renal disease: `renal`
- Viral hepatitis: `viral_hepatitis`
- HIV/AIDS: `hiv_aids`
- Substance-use or mental-health disorder: `substance_use_or_mental_health_disorder`
- Other accidental injury: `accidental_injury_other_than_drug_overdose`

Four modeled outcomes

Variable	Construction
`cardiovascular`	Copy of `cardiac_hypertension`.
`cerebrovascular`	`I60-I69` indicator described above.
`other_medical`	Equals 1 when any of cancer, diabetes/metabolic, hepatobiliary/pancreatic, renal, viral hepatitis, or HIV/AIDS equals 1.
`no_additional`	Equals 1 when cardiovascular, cerebrovascular, and other medical all equal 0.

The exact ICD-10 regular expressions are documented directly above `condition_patterns` in `scripts/01_build_analysis_data.R`.